Q1:JUN 2008 P2

3 A car travelling round a circular bend of radius 25 m in a road at a constant speed of 22 ms^{-1}.

(a) The velocity is not constant. Explain why.

____

____

____ [1]

(b) **Fig. 3.1** illustrates the position of the car at an instant during this motion.

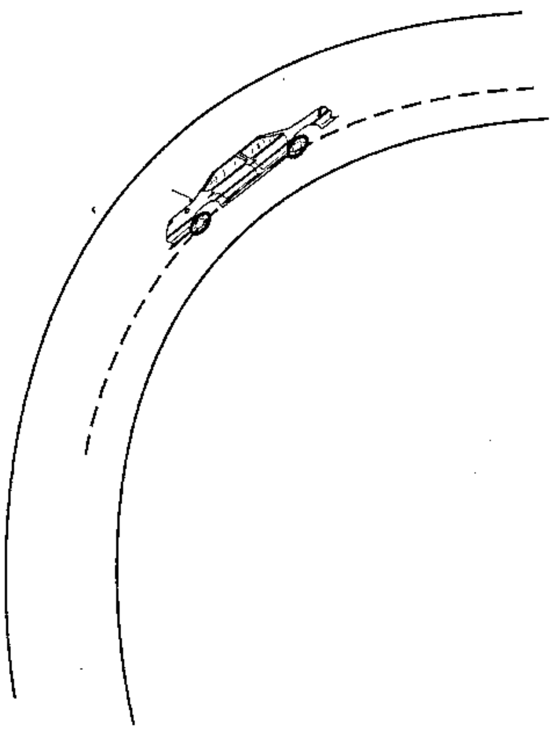

Fig. 3.1

(i) On **Fig. 3.1** show the directions of the velocity and the acceleration. [2]

(ii) Calculate the magnitude of the car's acceleration.

acceleration = ____________________ [2]

(iii) State the origin of the force which causes this acceleration.

____

____ [1]

Q2:NOV 2011 P2

2 **Fig. 2.1** shows a hammer thrower as he rotates with an angular speed of 2 revolutions per second just before releasing the hammer at an angle of $\theta = 30°$ to the horizontal. The head of the hammer, B, is 2.000 m above the ground.

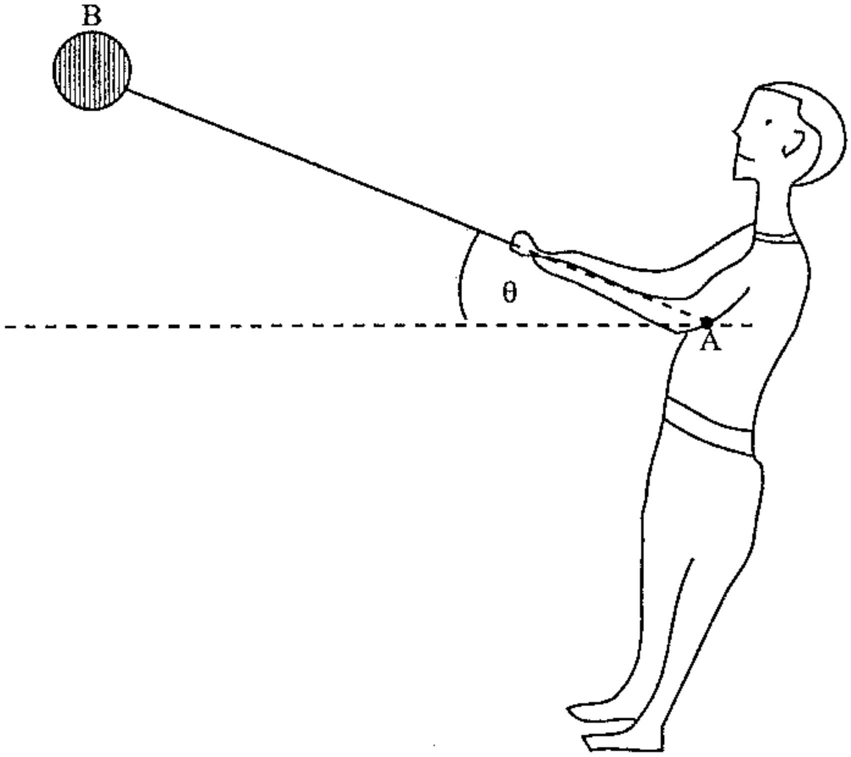

Fig. 2.1

The length from the centre of mass of B to A is 2 100 mm.

Calculate

(i) the angular speed, ω, of the hammer just before release,

$\omega =$ ______________________

(ii) the speed, v, of the hammer at release,

$v =$ ____________________

(iii) the time, t, taken for the hammer head, B, to reach the ground.

$t =$ ____________________

[7]

Q3:JUN 2016 P2

3 A car of mass 900 kg moves with a uniform speed of 15 ms^{-1} in a circular path of radius 90 m on a horizontal surface.

(i) State how the centripetal force, F, is produced.

____

____

(ii) Determine the magnitude and direction of F.

$F =$ ____________________ N

direction ____________________

____

(iii) Explain why roads are banked at the corners, considering all the forces acting on the car when it moves round a curve.

____

____

____

Q4:2018 SPEC P2

1 **(a)** Define *angular velocity*.

____

____ [1]

(b) A body is moving in a circular path of radius, r, at constant linear velocity, v, angular velocity, ω, and period T.

Deduce an expression connecting v, r and T.

____

____ [1]

(c) **Fig. 1.1** shows a 0.6 kg stone tied to one end of a string whirled in a vertical of circle of radius 0.4 m at a constant rate of 12.0 turns per minute.

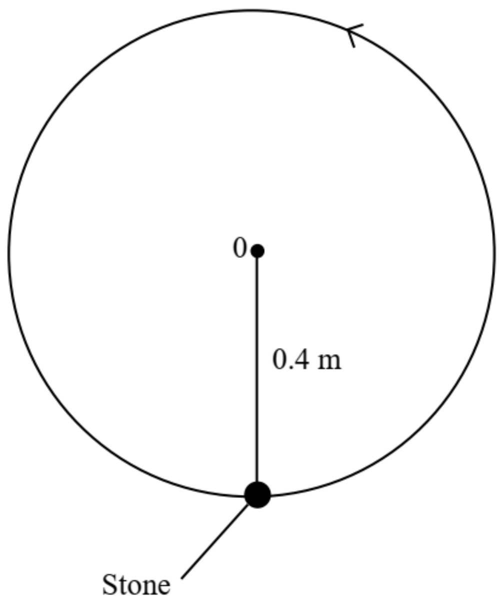

Fig. 1.1

(i) Show on **Fig. 1.1**, the direction of linear velocity v and centripetal acceleration, a.

(ii) Calculate the centripetal acceleration.

centripetal acceleration = _________

(iii) Label a point Q where the stone would be when the string is most likely to break.

____

____

(iv) Determine the tension in the string at Q.

Tension = ________ [6]

(d) Explain why passengers experience a normal reaction less than their weight when the vehicle goes over the top of a curved bridge.

____

____

____

____ [2]

Q5:JUN 2011 P3

1 **(b)** **(i)** A body with a constant speed in a uniform circle experiences an acceleration towards the centre of the circle.

Use Newton's laws to explain how this is possible.

(ii) **Fig. 1.1** shows a stone of mass, m, being whirled continuously in a vertical circle of radius r.

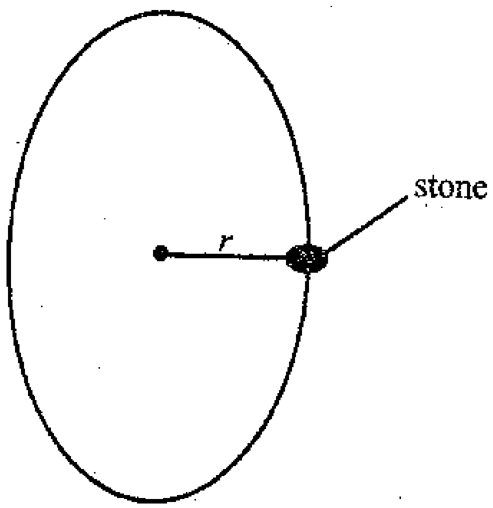

Fig. 1.1

State and explain when the tension in the string is

1. greatest,
2. lowest.

Q6:NOV 2011 P3

1 (a) **Fig.1.1** shows a car moving on top of a hill with a constant speed, v.

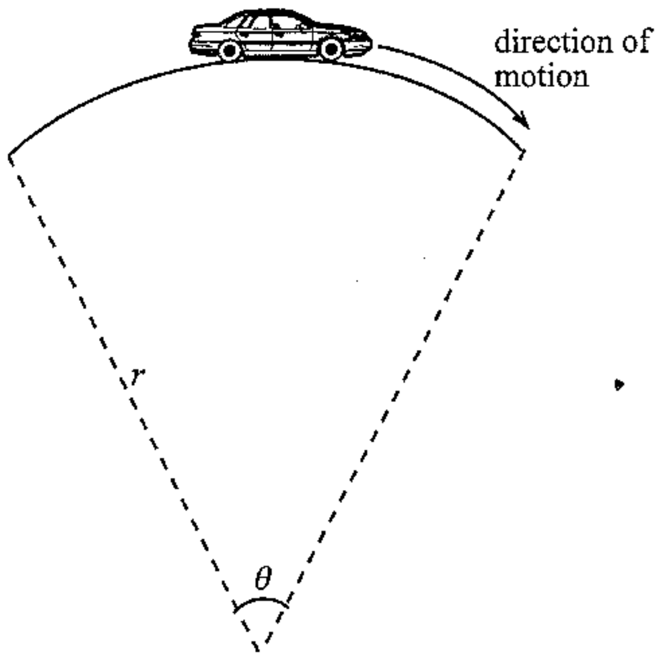

Fig. 1.1

(i) Show that the steady speed v of the car is given by $v = \omega r$ where ω is the angular velocity and r the radius of curvature.

(ii) Explain why the car experiences

1. a normal reaction less than its weight,

2. an acceleration. [5]

Q7:NOV 2015 P3

1 (c) **Fig.1.1** shows part of a car moving at 15 ms^{-1} on a level road. The wheel is of radius 0.30 m and has an angular velocity, ω, of 50 rad s^{-1}.

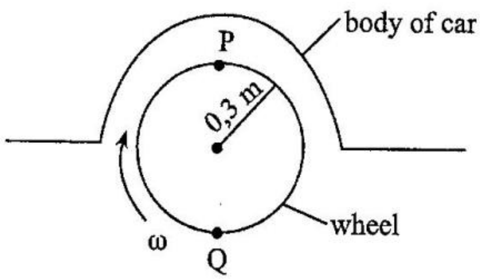

Fig.1.1

(i) Determine the minimum co-efficient of friction to avoid skidding given that the car is going round a curve of radius 200 m at 15 ms^{-1}.

(ii) Point P is observed as moving at twice the velocity of the centre of the wheel while point Q is observed as momentarily at rest.

Explain these observations.

[5]

Q8:JUN 2016 P3

3 (a) Define *centripetal force*. [2]

(b) Two popular attractions at an amusement park are the big wheel and the dangler which are shown in **Fig. 3.1 A** and **B** respectively.

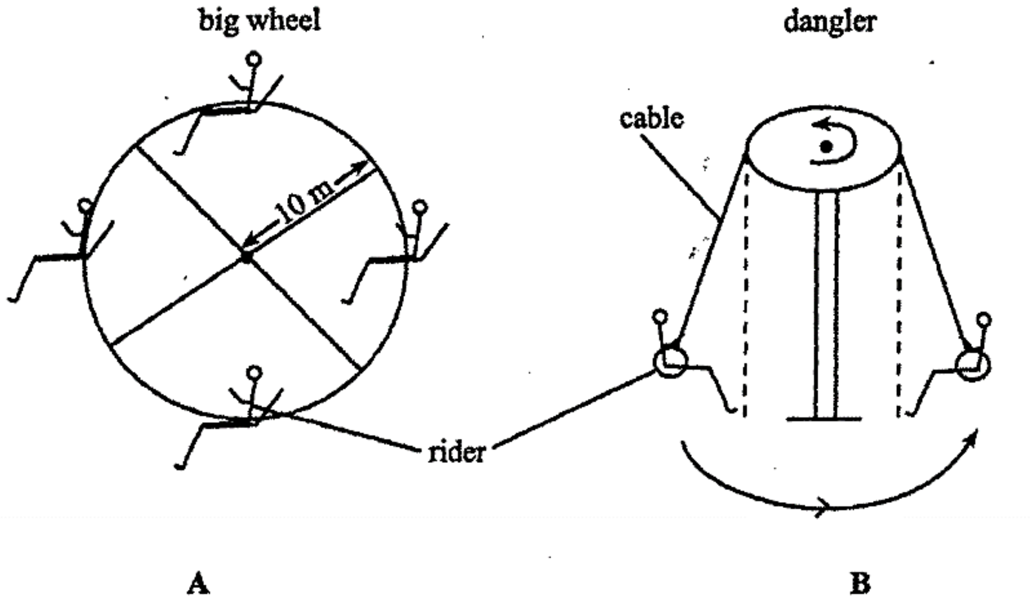

Fig. 3.1

The big wheel is made to rotate in a **vertical circle** while the dangler rotates in a **horizontal circle.**

(i) Explain each of the following observations:

1. when the dangler rotates the riders swing outwards and upwards

2. a rider on the big wheel feels heavier when at the bottom and lighter when at the top

(ii) Show that, for safety reasons, the rotational speed of the big wheel must not exceed 0.99 $rads^{-1}$.
[6]

(c) Describe and explain a possible danger of rotating the dangler at high speeds. [2]

Q9:JUN 2017 P3

1 (d) (i) Define *angular velocity*.

(ii) Derive the formula $v = r\omega$, where symbols have their usual meaning.

(iii) Explain why a particle moving with constant speed along a circular path has an acceleration.
[6]

Q10:JUN 2018 P3

1 (c) (i) State what provides the centripetal force for an aeroplane turning in flight.

(ii) **Fig.1.1** shows a 50 g metal ball secured to a ceiling by a weightless string of length 1.5 m. The ball is raised through a height of 50 cm and then released.

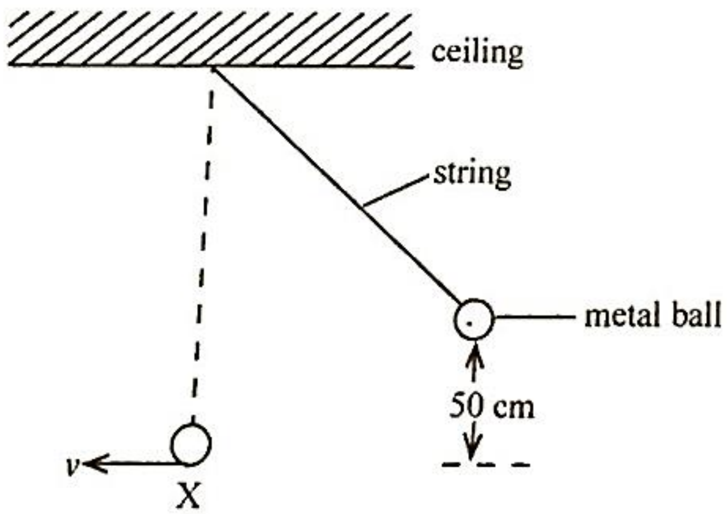

Fig.1.1

1. Calculate the speed, v, as the ball passes through point X, the equilibrium position.

   (ignore air resistance)

2. State and explain whether or not the tension in the string is equal to the weight of the ball at point X. [5]

Q11:N0V 2018 P3

1 (d) A grinding blade of diameter 40 m, rotates at 600 revolutions per minute when in use.

Calculate for a point on the edge of the blade, its

(i) linear speed,

(ii) acceleration. [4]

Q12: JUN 2019 P3

1 (d) (i) Define *centripetal force*.

(ii) A 0.8 kg object, attached to a string 0.5 m long, is whirled in a vertical circle at a constant speed. The minimum tension in the string is 30 N.

Calculate the

1. angular frequency,

2. linear speed,

3. maximum tension in the string.

[6]

Q13:NOV 2017 P3

(c) **Fig. 1.1** shows an object of mass m, tied at the end of an inextensible string rotated in a vertical circle of radius r, at a constant speed v and angular velocity ω. A and B are positions of the particle at velocities v_A and v_B respectively.

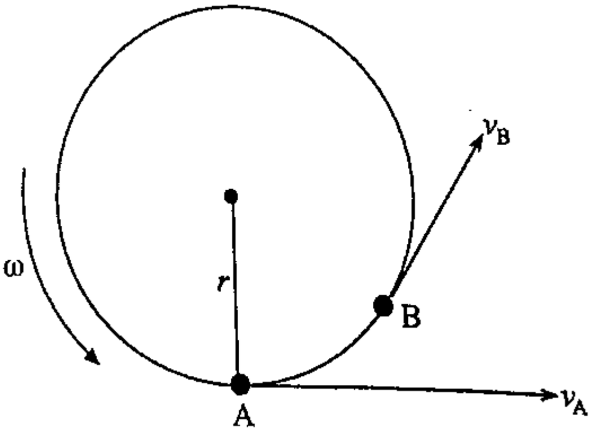

Fig. 1.1

The velocity changes from v_A to v_B and the resultant velocity Δv is given by the expression $\Delta v = v_B - v_A$.

(i) Draw a vector triangle of the velocity.

(ii) Explain why the object is in an accelerated motion when the speed v is constant. [4]

Q14:NOV 2019 P2

1 **(a)** **Fig. 1.1** shows a student whirling a can at the end of a string.

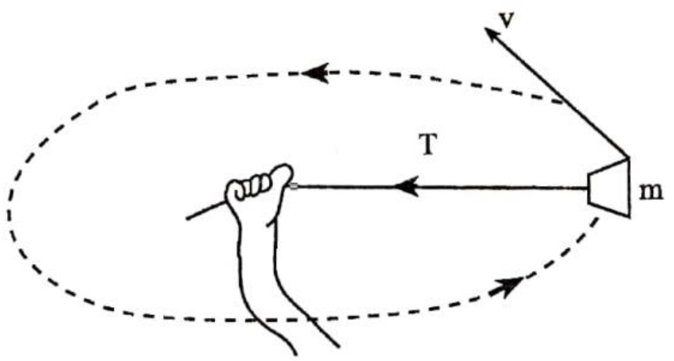

Fig. 1.1

Explain why the string cannot be held horizontal while the can is rotating in a circular path.

Q15:NOV 2019 P3

(e) **Fig. 1.4** shows a metal arm of a fairground ride that propels a toy rocket of mass 140 kg in a horizontal circle of radius 5 m with a constant speed of 8 ms^{-1}.

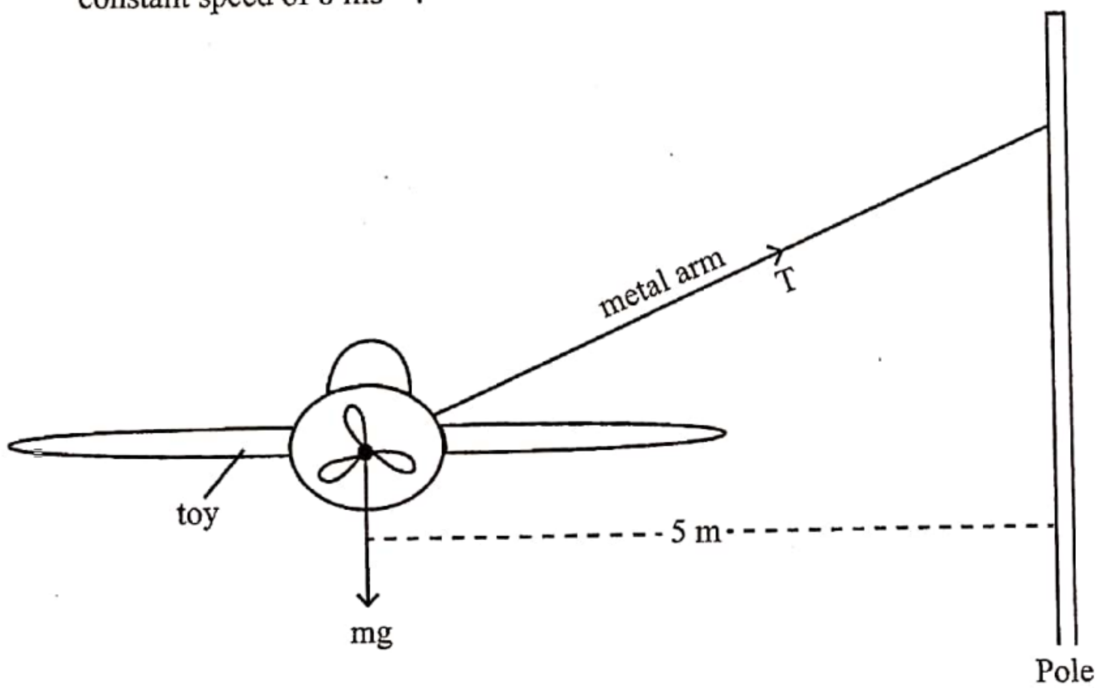

Fig. 1.4

If the metal arm is slightly inclined to the horizontal, find the tension, T, in the metal arm. [3]